

What we're doing

Use 50 iid contacts-v1 rollouts to infer stable contact-occurrence clusters, then allocate 50 branch rollouts equally across those clusters. Each branch is prompted with a coherent 3- or 5-contact bundle observed together in one warm-up map. Compare with iid100 and paired random coherent bundles.

Why

#321 improved oracle accuracy by strengthening samples but made them more similar. This experiment tests the complementary lever: spend part of the same 100-rollout budget on stable minority contact states. Single-contact seeding was too weak, while large predicted prompts can amplify errors, motivating small coherent bundles and a development gate.

Development gate

The $k=5$ policy passed on 16 frozen development proteins: continuation-only oracle fixed-R was 0.4893, +0.0094 versus iid100 and +0.0083 versus random $k=5$. The $k=3$ gain versus iid was only +0.0042, below the +0.005 gate. We froze $k=5$ before opening the other 81 eval-val proteins.

Held-out oracle: no replication

On 81 untouched proteins, cluster $k=5$ scored 0.5296 oracle fixed-R versus 0.5331 for iid100: delta -0.0035, 95% CI [-0.0078, +0.0009]. It also trailed random $k=5$ by -0.0023. The 39-protein subset with a real stable cluster was still flat-to-negative, so fallback dilution does not explain the failed replication.

Consensus: small generic seeding effect

Held-out all-range consensus increased +0.0036 versus iid100, but cluster $k=5$ was -0.0008 versus random $k=5$. Long-range consensus fell -0.0038 versus iid and -0.0053 versus random. The all-range increase came from fallback targets, not targets on which stable clusters existed.

Coverage is the bottleneck

The strict $k=5$ split-half criterion found an eligible cluster on only 8/16 development and 39/81 held-out proteins. $k=3$ had much better coverage but did not clear the oracle gate. Pairwise contact co-occurrence from 50 maps appears too noisy and local to identify reusable structural basins.

Fold-switch stress test is also null

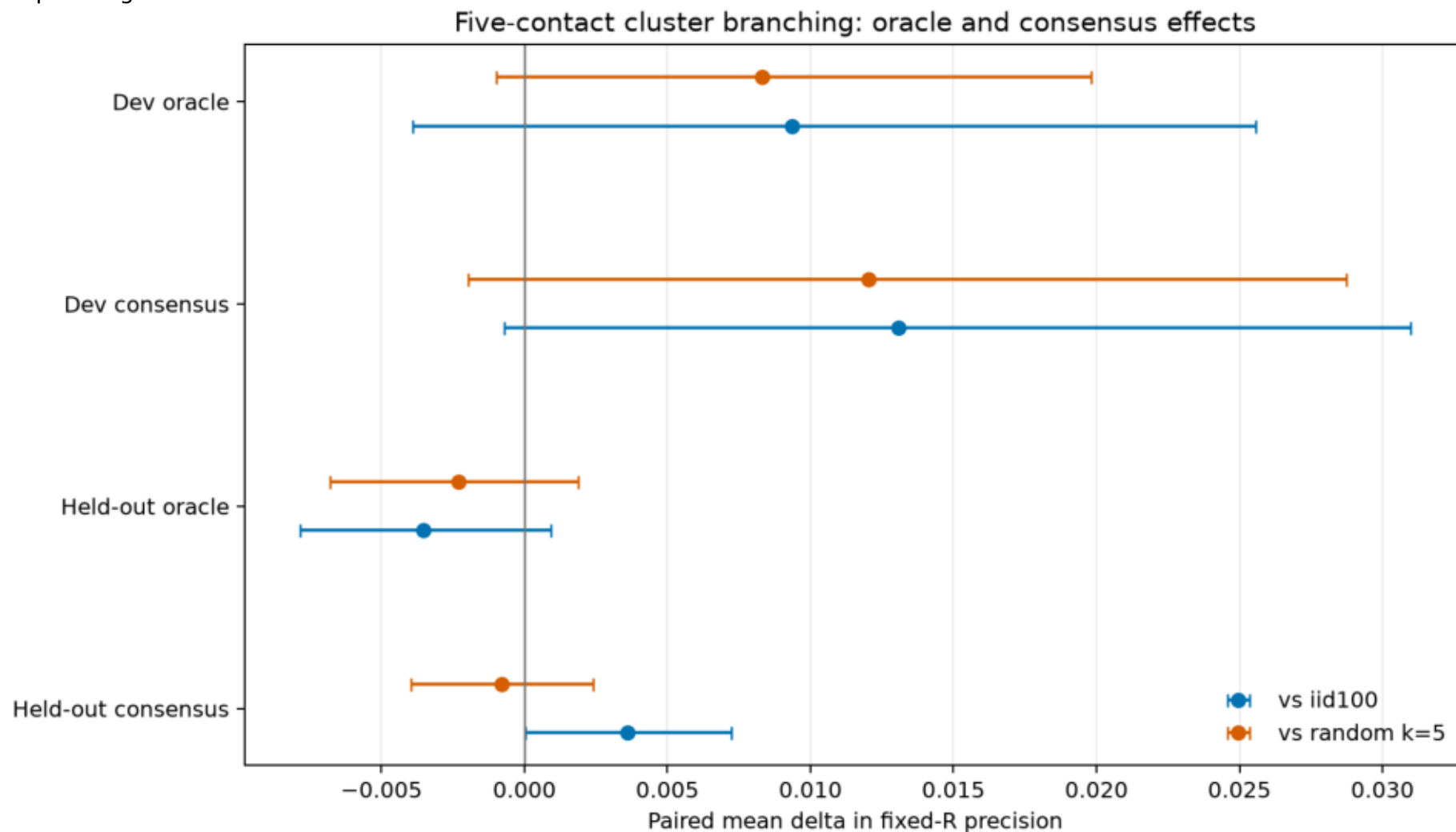
On 15 frozen fold-switch development pairs, iid100, random $k=5$, and cluster $k=5$ each reached fold 1 on 10 pairs, fold 2 on four, and both modes on one. Cluster $k=5$ had slightly higher pairwise Jaccard than iid100 (0.2782 vs 0.2771), despite eligible stable clusters on 12/15 pairs.

Conclusion

Do not adopt this brancher for oracle best-of-100 inference. The development gain was selection noise. A future attempt should cluster whole contact maps with a geometry-aware distance and branch from medoid partial maps, while retaining iid100 and same-prompt-size random controls.

01_paired_accuracy_deltas.png

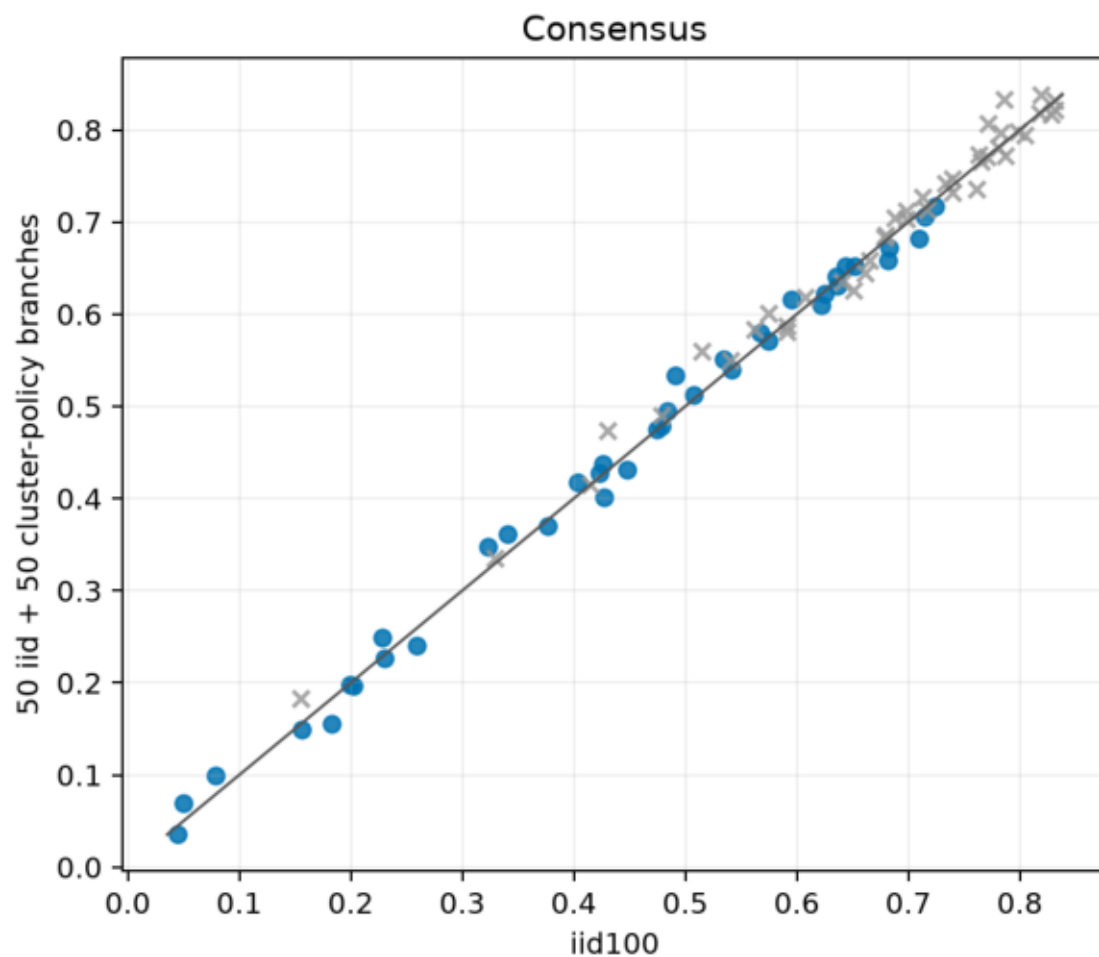
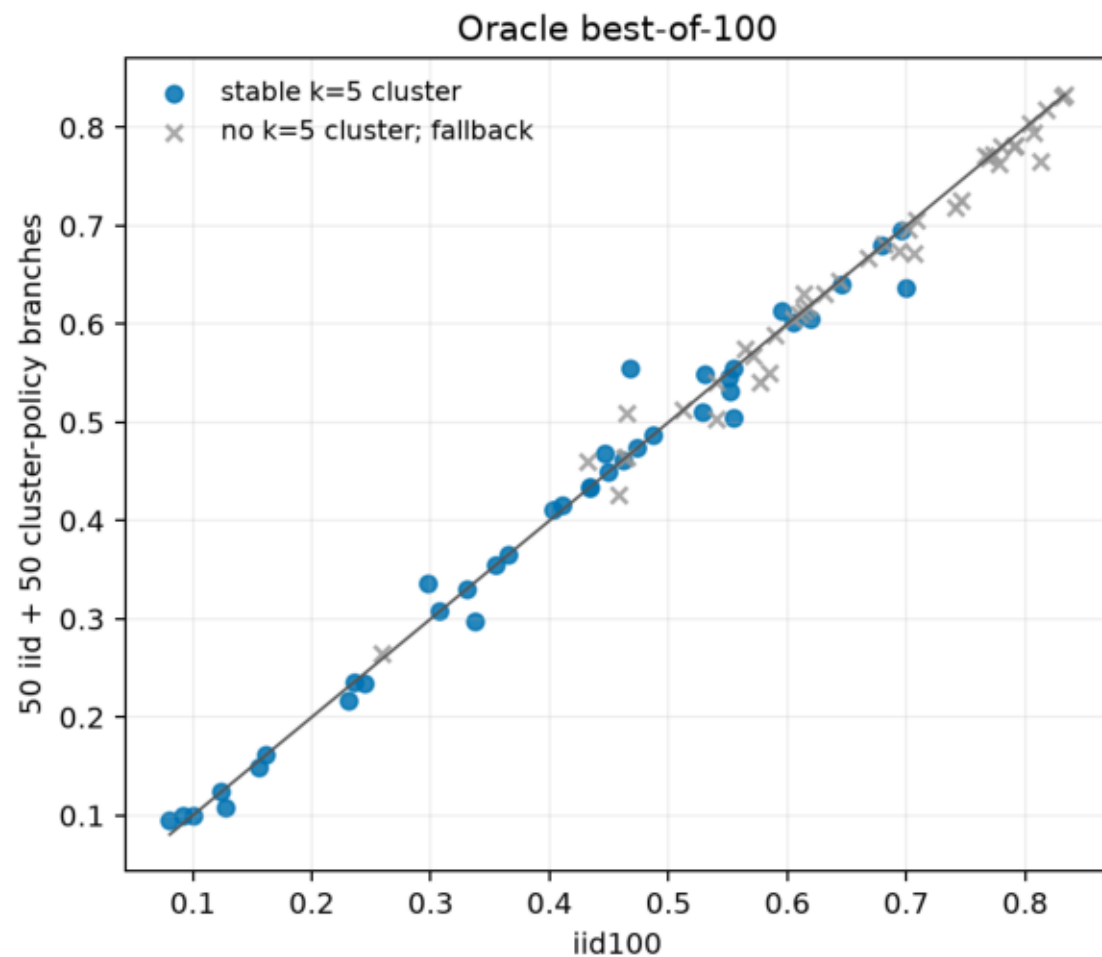
Protein-paired bootstrap mean differences for the primary continuation-only map. The held-out split was opened only after k=5 cleared the development gate.



02_heldout_per_protein.png

Each point is one of 81 held-out eval-val proteins. Blue points had at least one split-half-stable five-contact cluster; gray crosses used the preregistered coherent-random fallback.

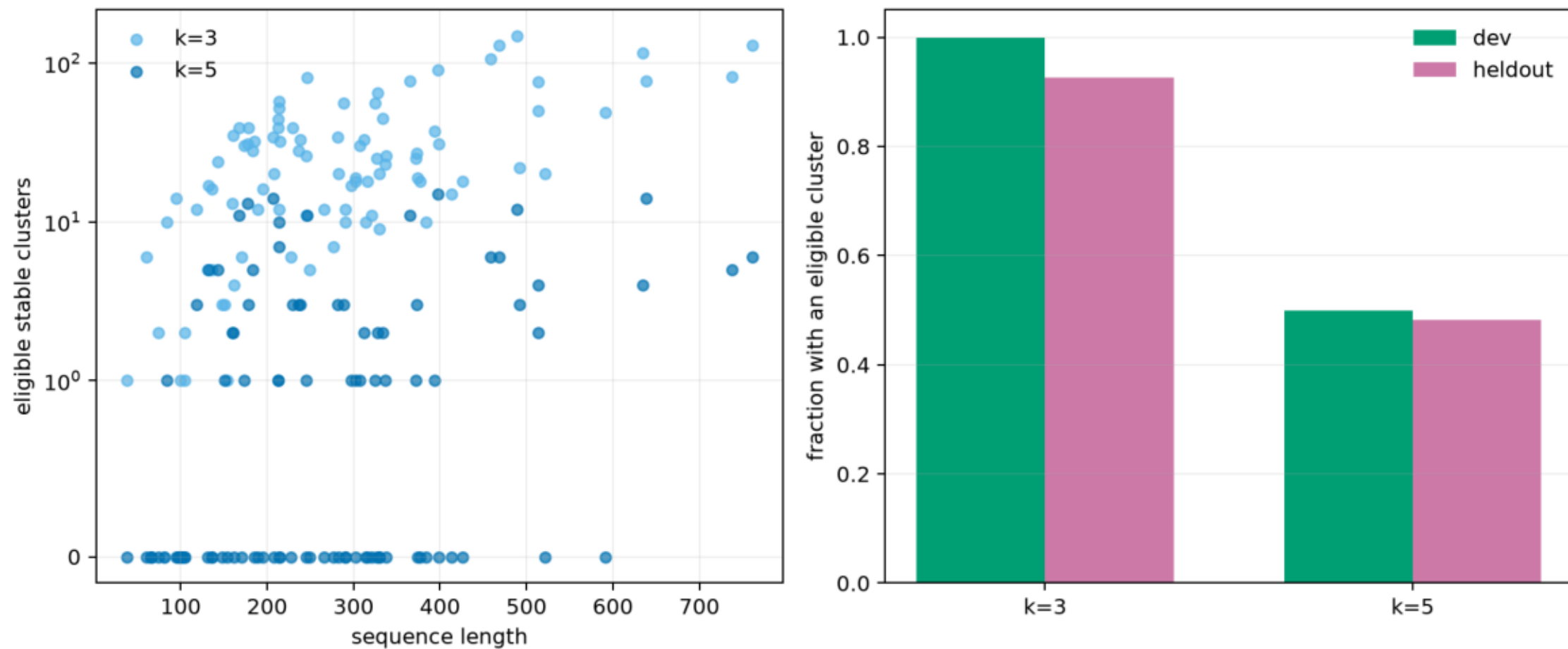
Frozen held-out split: per-protein effects



03_cluster_coverage.png

Three-contact clusters are broadly available; the stricter five-contact arm falls back on about half the proteins, an important limitation of the frozen policy.

Truth-free split-half cluster coverage



04_foldswitch_modes.png

Oracle contact-mode coverage on 15 frozen fold-switch development pairs, using the same total 100-map budget and excluding supplied contacts from branch continuations.

